

LM01 Fixed-Income Instrument Features

1. Introduction.....	2
2. Features of Fixed-Income Securities.....	2
3. Bond Indentures and Covenants	5
Summary.....	8

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1. Introduction

This learning module introduces the features of fixed income instruments and the legal contracts that govern them.

2. Features of Fixed-Income Securities

Fixed-income instruments, such as loans and bonds, are debt instruments that represent a contractual agreement in which an issuer borrows money from investors in exchange for interest and future repayment of principal.

Basic Features of a Bond

The basic features of a fixed-income security include:

Issuer:

Issuer is the entity that has issued the bond, or the one who has borrowed money. It is responsible for servicing the debt (paying interest and principal payments). Bonds can be classified into the following categories based on the type of the issuer:

- **Supranational organizations:** Bonds issued by international organizations such as the World Bank and IMF.
- **Sovereign governments:** Debt of national governments such as government of the United States, Germany, or Italy.
- **Non-sovereign governments:** Bonds that are not issued by a national or central government. Instead, these are issued by states, provinces, municipalities, etc. For example, municipal bonds in the United States, etc.
- **Quasi-government entities:** Bonds issued by agencies that are financed either directly or indirectly by the government. They are also called sub-sovereign or agency bonds. For example, Ginnie Mae, Fannie Mae, and Freddie Mac in United States.
- **Companies:** Bonds issued by a corporate. A distinction is often made between financial issuers and non-financial issuers.

All bonds are exposed to credit risk. **'Credit risk'** is the risk that interest and principal payments will not be made by the issuer as they come due. Credit rating agencies such as Moody's and S&P assign a rating to issuers based on this risk.

Bonds can be classified into two categories based on their creditworthiness:

- Investment grade.
- Non-investment grade, or high-yield or speculative bonds.

Maturity:

The maturity date is the date on which the issuer will redeem the bond by paying the outstanding principal amount. Once a bond has been issued, the time remaining until maturity is known as the tenor of the bond.

- If original maturity is one year or less, the bond is called money market security.

- If original maturity is more than a year, the bond is called capital market security.

Par value:

Also known as face value, maturity value, or redemption value.

- It is the principal amount that is repaid to bondholders at maturity.
- If market price > par value, the bond is trading at a premium.
- If market price < par value, the bond is trading at a discount.
- If market price = par value, the bond is trading at par.

Coupon rate and frequency:

Coupon rate is the percentage of par value that the issuer agrees to pay to the bondholder annually as interest. It can be a fixed rate or a floating rate. The coupon frequency may be annual, semi-annual, quarterly, or monthly. Based on the frequency of coupon payments, we can classify bonds into the following:

- *Plain vanilla bond*: It is the most basic type of bond with periodic fixed interest payments during the bond's life and the principal paid on maturity.
- *Floating-rate bond or floating-rate notes or floaters*: Unlike vanilla bonds, the interest rate of a floating-rate bond is not fixed. The coupon payments are based on a floating Market reference rate (MRR) at the start of the period. Some bonds specify the coupon rate as two components: a market reference rate, plus a spread. The coupon rate and coupon interest paid in every period change as the reference rate changes.
- Zero-coupon bonds or pure discount bonds: Bonds that have only one payment at maturity. These are bonds that do not make a coupon payment over a bond's life, and are sold at a discount (less than the par value) at issuance. At maturity, the investor receives the par value of the bond. Think of it as interest getting accumulated during the bond's life and being paid at maturity for a zero-coupon bond.

Seniority:

A single borrower may issue bonds with different seniority rankings. Seniority ranking determines who gets paid first, or who has the first claim on the cash flows of the issuer, in the event of default/bankruptcy/restructuring. Senior debt has a priority over other debt claims. Junior or subordinated debt has a lower priority and is paid only once the senior claim are satisfied.

Contingency provisions:

A contingency provision is a clause in a legal document that allows for some action if the event or circumstance does occur. It is also called an embedded option. The most common contingency provisions for bonds are: call, put, and conversion options.

Yield Measures

There are two widely used yield measures to describe a bond: current yield and yield to

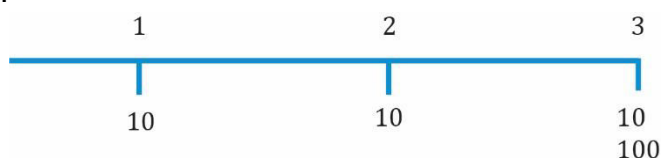
maturity.

Current yield or Running yield

Current yield is the annual coupon divided by the bond's price and is expressed as a percentage. For example, consider a three-year, annual coupon bond with a par value of \$100. The bond is issued at \$95. The coupon rate is 10%, so the coupon payments are \$10 every year. Current yield = $\frac{10}{95} = 10.5\%$

Yield to Maturity (YTM)

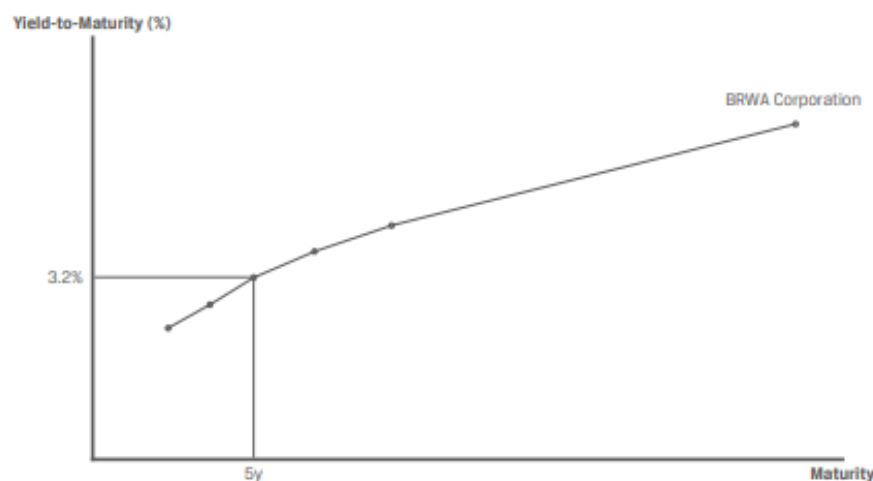
Yield to maturity is the internal rate of return on a bond's expected cash flows. In other words, it is the expected annual rate of return an investor will earn if the bond is held to maturity. It is also known as 'yield to redemption' or 'redemption yield'. The IRR can be calculated easily using the financial calculator. Let us consider the same bond as in the current yield example:



Input these values: $CF_0 = -95$; $CF_1 = 10$; $CF_2 = 10$; $CF_3 = 110$. Computing for IRR, you should get 12.08%. As you can see, YTM is higher than the coupon rate. For a discount bond such as this one, the YTM is higher than the coupon rate. A bond's price is inversely related to its yield to maturity.

Yield Curves

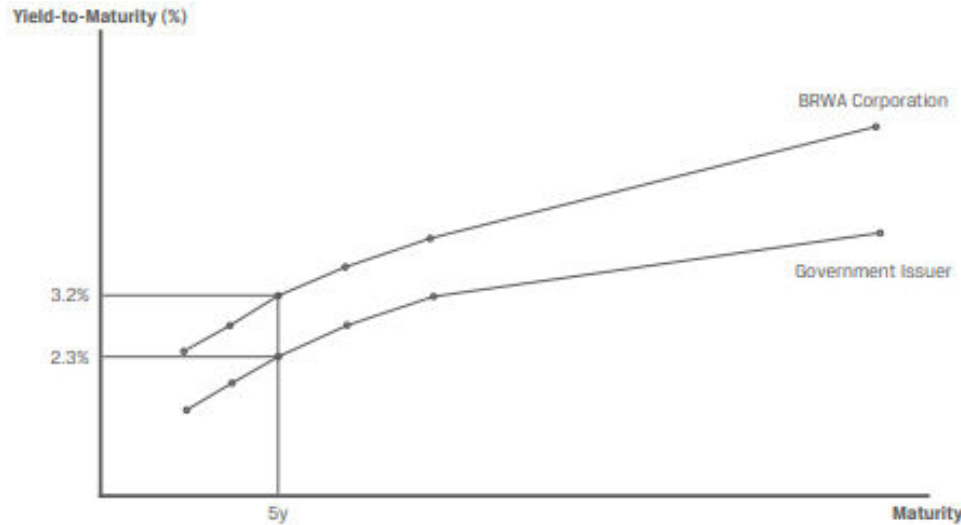
The yield curve plots the YTM's of bonds on the y-axis versus maturity on the x-axis. Exhibit 4 from the curriculum shows a sample yield curve. This curve is constructed using six bonds issued by a single corporation.



As seen in the graph, bonds with longer maturities have higher YTM's, indicating that investors are demanding higher expected returns to compensate for higher risk associated

with longer-maturity bonds.

To evaluate credit risk, we can compare an issuer's yield curve with the yield curve for comparable sovereign bonds (which have no credit risk). Exhibit 5 from the curriculum illustrates this.



The yield difference between the 5-year corporation bond and the 5-year US treasury bond is $3.2\% - 2.3\% = 0.9\%$. This reflects the compensation that investors demand for taking the additional credit risk associated with the corporation bond.

3. Bond Indentures and Covenants

Bond Indenture

A bond indenture is a written legal agreement between the bond issuer and the investor. The indenture has the following information:

- The name of the issuer.
- All the terms of a bond issue such as the type of bond.
- Its features such as the principal value, coupon rate, dates when interest payments will be made, and maturity date.
- Issuer's obligations.
- Bondholders' rights.
- If the bonds are secured or not.
- Covenants.
- Contingency provisions like call option.

Source of Repayment Proceeds

Investors need to know how the issuer intends to make interest payments and repay the principal. The bond indenture specifies the source of revenue or how the issuer intends to make interest and principal payments. We look at the repayment sources for each category

now.

- For supranational bonds: From repayment of previous loans or paid-in capital from its members. For example, the World Bank may have given a loan to a particular country. When the country pays back the loan, this money is used to pay the bondholders. Paid-in capital is the amount contributed by member nations.
- For sovereign bonds: Tax revenues and printing new money. The government may also increase taxes to service debt.
- Three sources for non-sovereign government debt issues: the taxing authority of the issuer, cash flows of the project, special taxes or fees.
- For corporate bonds: Cash flow generated by the company's primary operations. The higher the credit risk, the higher the yield.
- For securitized bonds: Cash flows generated by the underlying assets such as mortgages, credit card receivables, auto loans, etc. Securitized bonds are amortized, i.e., the principal is paid back gradually over the bond's life rather than in one payment at maturity.

Asset or Collateral Backing

Collateral reduces credit risk.

- Secured vs. unsecured bonds: In a secured bond, assets are pledged as collateral to ensure debt repayment in the case of a default. Unsecured bonds have no collateral.
- Seniority ranking: What this means is that if a company faces bankruptcy and its assets are liquidated, the investors (or bondholders) are repaid based on seniority.

Covenants

Bond covenants are legally enforceable rules that borrowers and lenders agree on at the time of a new bond issue. Covenants can be affirmative (positive) or negative (restrictive). Affirmative covenants indicate what the issuer must do. These are generally administrative in nature. They do not impose additional costs on the issuer. For example, the issuer must:

- Make interest and principal payments on time.
- Comply with all laws and regulations.
- Maintain current lines of business.
- Insure and maintain assets.
- Pay taxes on time.
- Other examples include a 'pari passu' clause which ensures that a debt obligation is treated the same as the borrower's other senior debt instruments, or a 'cross-default' clause which specifies that a borrower is considered in default if they default on another debt obligation.

Negative covenants indicate what the issuer must not do. These are frequently costlier and materially constrain the issuer's potential business decisions. Examples include:

- Restrictions on debt: Limits on maximum acceptable leverage ratios and minimum acceptable interest coverage ratios.

- Negative pledges: No additional debt can be issued that is senior to existing debt.
- Restrictions on prior claims: The issuer cannot collateralize assets that were previously collateralized. The objective is to protect unsecured bondholders.
- Restrictions on distribution to shareholders: Restrictions on how much money can be spent on dividends and buy-backs.
- Restrictions on asset disposals: A limit on the total amount of assets that can be disposed during a bond's life. The objective is to ensure the company does not breakup.
- Restrictions on investments: This ensures no investment is made in a speculative business and that the capital is invested in a going-concern business.
- Restrictions on mergers and acquisitions.

Summary

LO: Describe the features of a fixed-income security.

The basic features of a fixed-income security include the specification of:

- Issuer: It is the entity that has issued the bond. The types of issuers include supranational organizations, sovereign governments, non-sovereign governments, quasi-government entities, and companies.
- Maturity: It is the date on which the last payment is made for a bond. Based on maturity, bonds can be classified into money market security, capital market security, and perpetual bonds.
- Par value: It is the amount an issuer agrees to pay the investor or bondholder on maturity date. A bond is premium if sold above par, discount if sold below par, and at par if sold at face value.
- Coupon rate and frequency: Coupon rate is the interest rate paid on a bond every year by the issuer until its maturity date. Bonds that have only one payment at maturity are called zero-coupon bonds.
- Seniority: Seniority ranking determines who gets paid first, or who has the first claim on the cash flows of the issuer, in the event of default/bankruptcy/restructuring.
- Contingency provisions: Some bonds may be issued with embedded call/put/conversion options.

LO: Describe the contents of a bond indenture and contrast affirmative and negative covenants.

Bond indenture is a legal agreement between the bond issuer and the investor. It includes all the terms of a bond issue such as the type of bond, its features such as principal value and coupon rate, and maturity date. It also includes the dollar amount of issue, issuer's obligations, bondholders' rights, when the coupon payments will be made, if the bonds are secured or not, covenants, and contingency provisions.

Bond covenants are legally enforceable rules that borrowers and lenders agree on at the time of a new bond issue. Covenants can be affirmative (positive) or negative (restrictive).

Affirmative covenants indicate what the issuer must do. These are generally administrative in nature. For example, issuer must make interest and principal payments on time, comply with laws and regulations, insure and maintain assets, etc.

Negative covenants indicate what the issuer must not do. These are frequently more costly and materially constrain the issuer's potential business decisions. Examples include restriction on debt, negative pledges, restrictions on prior claims, restrictions on distribution to shareholders, etc.